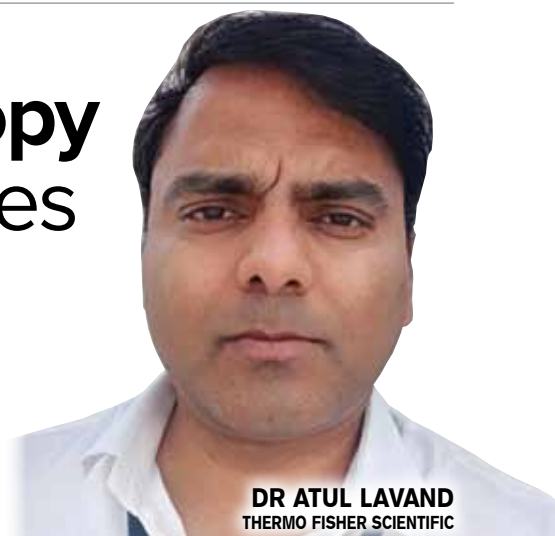


Test

“Molecular Spectroscopy Is Applicable To All Types Of PCBs”

Dr Atul Lavand, Application Product Specialist at Thermo Fisher Scientific, observes that favourable government policies are drawing foreign PCB manufacturers to India—driving demand for molecular spectroscopy. In conversation with EFY’s Yashasvini Razdan, he discusses its role in PCB failure analysis and material identification.



DR ATUL LAVAND
THERMO FISHER SCIENTIFIC

How is molecular spectroscopy used in the semiconductor and electronics industry?

In electronics, the primary focus is on research and development (R&D) teams, especially those working in PCB manufacturing. Molecular spectroscopy plays a crucial role in failure analysis. If a PCB fails, our Fourier transform infrared (FTIR) microscope can map and analyse specific areas of the board. FTIR light is directed onto the targeted region, producing a spectrum. This is then matched with a reference library to pinpoint the material responsible for the failure. Once identified, manufacturers can refine their processes to avoid similar issues in future.

Our FTIR microscopes are now in use with major electronics manufacturers, helping them pinpoint failure causes and improve production processes.

Are there other spectroscopy techniques or devices used in molecular spectroscopy?

Yes. Alongside FTIR spectroscopy, we use Raman spectroscopy and Raman microscopy. Both techniques complement each other—FTIR is preferred for identifying organic functional groups, while Raman is more

effective for inorganic ones.

Molecular spectroscopy is applicable to all types of PCBs. Whenever failure analysis is required, the FTIR microscope can be used to identify the material causing the issue.

How does molecular spectroscopy work in analysing multiple layers of coatings in PCBs?

The process is straightforward. When a failure occurs, the PCB is placed under the FTIR microscope. The specific area responsible for the failure is mapped, and FTIR light is directed onto it. The light interacts with the material, and the response is detected using a mercury cadmium telluride (MCT) detector. The resulting spectrum is compared with a global reference library to identify the material. Once identified, manufacturers can trace its source in the production process and make necessary adjustments to prevent defects.

What if there are any contaminants or residue in between the coatings?

Yes, molecular spectroscopy can detect that as well. Contaminants have unique spectral signatures. Molecular spectroscopy helps trace their origin, whether from packaging or the production process, enabling corrective actions.

India has a limited number of PCB manufacturers, with most companies focusing on assembly. What kind of demand do you see for molecular spectroscopy in this space?

You are right; PCB manufacturing in India is still developing. However, several foreign companies are establishing manufacturing units here due to favourable policies. These companies are our primary targets.

With proposals to set up outsourced semiconductor assembly and test (OSAT) units in India, the semiconductor industry is projected to grow. This growth will likely fuel increased demand for molecular spectroscopy.

Does artificial intelligence (AI) play any role in molecular spectroscopy?

AI is expected to play a significant role in the coming years, particularly in software for spectral interpretation. Thermo Fisher is developing AI-powered molecular spectroscopy software, which may become available soon. As AI capabilities mature, spectral interpretation is poised to become faster and more accurate. Thermo Fisher is already developing tools to lead this shift. **EFY**

This interview is an excerpt of a longer discussion. For the full transcript, please visit: <https://efy.co.in/efyjune25>

